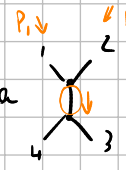
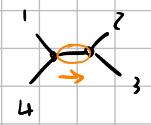
$A_4 = a$  $+ b$  generic A_4 within our ansatz

$$= \frac{a}{(P_1+P_2)^2} + \frac{b}{(P_1+P_4)^4} = \frac{a}{X_{13}} + \frac{b}{X_{42}} = \frac{1}{X_{13}}(a-b)$$

impose zero condition: $S_{13} = 0 \iff 0 = X_{13} + X_{24} - \cancel{X_{23}}^{\,0} - \cancel{X_{14}}^{\,0}$

$\downarrow$

if $X_{13} = -X_{24}$ then $A_4 = 0$ (assumption)

<u>recursion relation</u> $\quad c_{ij} = X_{ij} + X_{i+1\,j+1} - X_{i+1\,j} - X_{i\,j+1}$

$$c_{13} = c_{14} = \cdots = c_{1\,n-1} = 0 \iff \left.\begin{aligned} X_{24} &= X_{14} - X_{13} \\ X_{25} &= X_{15} - X_{13} \\ &\vdots \\ X_{2n} &= X_{1n} - X_{13} = -X_{13} \end{aligned}\right\} \; n-3 \text{ constraints}$$

Conversely, if we shift $\quad P_2 \to P_2 + qz$

$\qquad\qquad\qquad\qquad P_n \to P_n - qz$

the only affected X_{ij} are the ones that contain only 1 of P_2 or P_n

$X_{1j} \quad j = 3, \dots, n-1$

$X_{2j} \quad j = 3, \dots, n$

$$X_{1j}(z) = X_{1j}(0) + z\, X_{1j}^{\infty}$$

$$X_{1j}(z) = (\underbrace{P_1+P_2}+\underbrace{qz}+\underbrace{\cdots+P_{j-1}})^2 = \cancel{z^2q^2}^{\,0} + X_{1j}(0) + \underbrace{2zq\cdot(P_1+P_2+\cdots+P_{j-1})}_{X_{1j}^{\infty}}$$

$$X_{2j}(z) = X_{2j}(0) + \underbrace{2zq\cdot(P_2+P_3+\cdots P_{j-1})}_{X_{2j}^{\infty}}$$

$$= X_{2j}(0) + X_{1j}(z) - X_{1j}(0) - \underbrace{2zq\cdot P_1}_{X_{13}^{\infty}}$$

$$X_{13}^{\infty} = 2q\cdot(P_1+P_2) = 2q\cdot P_1$$

$$X^{\infty}_{2j} = X^{\infty}_{1j} - X^{\infty}_{13}$$

<u>Subset zeros</u> B is an homogeneous rational function of the X_{ij}

collect B into subterms depending on how they scale under the shift

$P_2 \to P_2 + zq \qquad P_n \to P_n - zq$

$B(z) = B_m + B_{m-1} + B_{m-2} + \cdots$

where B_m contains all terms that scale like z^m (as $z \to \infty$)

<u>Prop.</u> these are equivalent:

- B satisfies 1-zero
- each B_i satisfies 1-zero
- each B_i has enhanced <u>UV scaling</u> z^{i-1}

obvious

<u>step 1</u> if B satisfies 1-zero then it must have enhanced scaling.

B satisfies zero <u>means</u>: if $X_{2j} = X_{1j} - X_{13}$ then $B = 0$ $\forall$ other choices of X_{ij}

1-zero constraint

$j = 4, \ldots n$

$\to$ B must be at least linear in $c_{13}, c_{14}, c_{15}, \ldots$

$$B = \sum_{j=3}^{n-1} c_{1j} \{ \cdots \}_j$$

<u>ex!</u> $B = \dfrac{1}{X_{13}} + \dfrac{1}{X_{24}}$

$$0 = -c_{13} = X_{13} + X_{24} - \cancel{X_{14}} - \cancel{X_{23}}$$

$$= \frac{X_{24} + X_{13}}{X_{13}\, X_{24}}$$

what happens to B as $z \to \infty$

$$B \to z^m B_m(X_{ij}^\infty) + O(z^{m-1})$$

ex: $B = \frac{1}{X_{13}} + \frac{1}{X_{24}}$

$X_{24}(z) = X_{24}(0) + z X_{24}^\infty$

$X_{13}(z) = X_{13}(0) + z X_{13}^\infty$

$X_{24}^\infty = - X_{13}^\infty$

$B = B_{-1}$

$$B(z) = \frac{1}{X_{13}(0) + X_{13}^\infty z} + \frac{1}{X_{24}(0) - X_{13}^\infty z} = \frac{X_{13}(0) + X_{24}(0) + \cancel{X_{13}^\infty z} - \cancel{X_{13}^\infty z}}{-X_{13}^\infty z^2 + \dots} \to \frac{1}{z^2}$$

$$= \frac{1}{X_{13}^\infty z} + O(z^{-2}) \qquad \to -\frac{1}{X_{13}^\infty z} + O(z^{-2})$$

B has enhanced scaling $\to \left.\frac{1}{X_{13}}\right|_{X_{13} = X_{13}^\infty} + O(z^{-2})$

$f(z) = z$

$g(z) = -z$

$f + g = 0$

$$B = \frac{1}{1+z} + \frac{1}{2-z} = \frac{2 - z + 1 + z}{(1+z)(2-z)} = \frac{3}{\dots z^2}$$

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expand series at ∞?

$f(z) = \frac{1}{a + bz}$ expand at ∞

change of variable $z = 1/w$

expand at $w = 0$

$$f = \frac{1}{a + b/w} = \frac{w}{wa + b} = \frac{w}{b} + O(w^2) \quad \text{(taylor series)}$$

$$f'(0) = \left.\frac{(wa+b) - wb}{(wa+b)^2}\right|_{w=0} = \frac{b}{b^2} = \frac{1}{b}$$

$$\hookrightarrow \frac{1}{bz} + O(z^{-2})$$

---

as $z \to \infty$ $\quad B_m(z) \to z^m B_m(X_{ij}^\infty) + O(z^{m-1})$ $\quad$ Taylor series around $z = \infty$

B_m has enhanced scaling $\Leftrightarrow B_m(X_{ij}^\infty) = 0 \quad \forall X_{ij}^\infty$

X_{ij}^∞ satisfies the same relations as 1-zero

$B_m(X_{ij}^\infty) = 0 \quad \forall X_{ij}^\infty \iff B_m$ satisfies 1-zero

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this proves that ② $\iff$ ③ and ② $\Rightarrow$ ①

suppose ① is true

want to show

B satisfies 1-zero $\Rightarrow$ B_m satisfies 1-zero

if we let $X_{2i}(0) = X_{1i}(0) - X_{13}(0)$ then $B = 0$

then we do shift then $X_{2i}(z) = \overbrace{X_{1i}(z)}^{X_{1i}(0) + z X_{1i}^\infty} - X_{13}(z)$

$\downarrow$

$X_{2i}(0) + z X_{2i}^\infty$

then $\overset{B(X_{i}(z))}{B(z)} \to 0$ as $z \to \infty$

since $B \Rightarrow z^n B_m(X_{ij}^\infty) + O(z^{m-1})$ it must be $B_m(X_{ij}^\infty) = 0$

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let $B' = B - B_m = B_{m-1} + \cdots$

ex: $B(x,y) = x - y$

$x = 1 + z$

$y = -1 - z$

$B(1+z, -1-z) = 0$